OS2-1

Babbling opens the sensory phase for imitative vocal learning

Albertine Leitão, Manfred Gahr

Affiliation of presenting author:

Max Planck Institute for Biological Intelligence, 82319 Seewiesen, Germany

Zebra finches, a species of songbirds, learn to sing by creating an auditory template through the memorization of model songs (sensory learning phase) and subsequently translating these perceptual memories into motor skills (sensorimotor learning phase). It has been traditionally believed that babbling in juvenile birds initiates the sensorimotor phase while the sensory phase of song learning precedes the onset of babbling. However, our findings challenge this notion by demonstrating that testosterone-induced premature babbling actually triggers the onset of the sensory learning phase instead. We reveal that juvenile birds must engage in babbling and self-listening to acquire the tutor song as the template. Notably, the sensory learning of the template in songbirds requires motor vocal activity, reflecting the observation that prelinguistic babbling in humans plays a crucial role in auditorylearning for language acquisition.

OS2-2

Elephant vocal learning: Mechanisms, flexibility, and cognitive insights

Angela Stoeger, Anton Baotic

Affiliation of presenting author:

Acoustic Research Institute, Austrian Academy of Sciences

A central question in cognitive biology is whether the mechanisms underpinning human speech—such as vocal production learning—are shared with nonhuman animals. Elephants are now recognized as vocal learners, supported by growing evidence for heterospecific imitation and invented, altricial sounds. They possess control over multiple sound production systems, including laryngeal and synapomorphic sound sources, making them particularly suited for exploring complex vocal behavior. We report on adult African elephants (n = 13) trained to vocalize on verbal cues. Individuals reliably produced both natural social calls and novel, idiosyncratic sounds using diverse sound production strategies, including nasal tissue vibration via ingressive airflow and muscle contractions at the trunk base. Moreover, each elephant employed individualized techniques, indicating creative problem-solving and fine motor control. These findings demonstrate that elephants possess a broad vocal learning phenotype, combining production and usage learning with motoric creativity. Social feedback and positive reinforcement appear to support the development of novel vocal behaviors. Elephants offer valuable comparative insights into the evolution of flexible vocal communication and the cognitive foundations of language.

OS2-3

Sound production in amphibious Gobiidae: from water to land

Louise Eche, Eric Parmentier

Affiliation of presenting author:

Laboratory of Functional and Evolutionary Morphology, Department of Biology, Ecology, Evolution, University of Liège, Belgium.

The conquest of terrestrial environments required substantial morphological and physiological modifications for all lineages that undertook this transition to adapt to the new challenges posed by terrestrial life. While research has predominantly focused on Sarcopterygians, various species of Actinopterygians have also more or less successfully adapted to terrestrial habitats. To explore the modifications associated with environmental transitions, the Gobiidae family serves as an excellent model because some of its extant representatives, called mudskippers (Oxudercinae) inhabit both aquatic and terrestrial environments. This study aims to understand the evolution of acoustic communication during terrestrialization. We first recorded the sounds emitted by mudskippers underwater and on land. Simultaneously, the use of a camera allowed the analysis of movements during sound production and the identification of structures that are involved in acoustic communication. We identified different types of sounds that are produced both on land and underwater, with the pectoral girdle being the central structure involved in sound production. Through 3D reconstructions and histological observations, we gained detailed insights into the characteristics of the musculoskeletal system involved in the acoustic communication of these species. Studying exclusively aquatic gobies was required to make comparisons. Our results are a first step towards understanding the adaptations necessary for the persistence of acoustic communication following the transition to terrestrial environments.

OS2-4

How a long-range sonar is used for short-range echolocation: Sperm whales use quiet and frequent echolocation and swim upside-down when searching for prey close to the seafloor

Pernille Tønnesen, Olivia Marín Delgado, Mark Peter Johnson, Natacha Aguilar de Soto, Peter Teglberg Madsen

Affiliation of presenting author:

Department of Zoophysiology, Aarhus University, Denmark, Project CETI, New York, USA

All animals need to acquire food to survive. In the deep ocean, sperm whales locate prey by emitting the most powerful sound in the animal kingdom for long-range echolocation. Occasionally, sperm whales switch from pelagic to benthic-benthopelagic foraging near the highly reflective seafloor, raising the question of how they operate this powerful biosonar system in such a reverberant environment. We hypothesized that sperm whales foraging near the seafloor adjust their biosonar by 1) using faster click rates to match a shorter acoustic scene limited by the seafloor and by 2) using weaker output levels to reduce strong seafloor echoes. We used data from sound and movement recording DTAGs deployed on seven sperm whales in the Mediterranean Sea and the Gulf of Mexico. All whales performed both benthic and pelagic foraging. During steep descents toward the prey, the whales clicked twice as fast and used medians of 2 – 9 dB lower apparent output levels during benthic dives compared to pelagic dives. No obvious biosonar adjustments were observed during the foraging phase, where the whales are closest to the seafloor. During this phase, five of seven whales spent markedly more time swimming upside-down in benthic dives compared to pelagic dives. In this position the body of the whales likely blocks intense seafloor echoes from reaching the hearing organ in the lower jaw. We conclude that sperm whales employ contextual adjustments of their biosonar sampling rate, acoustic gaze, and body orientation to enhance prey search and capture in different foraging habitats.

OS2-5

Beyond speech: Investigating diversity in human vocal expression

David Reby, Valentina Canessa-Pollard, Katarzyna Pisanski, Mathilde Massenet and Andrey Anikin

Affiliation of presenting author:

ENES Bioacoustics Research Lab, CRNL, University of Saint-Etienne, France

Humans have developed voluntary control over vocal production for speech and singing, while retaining an ancient system for spontaneous nonverbal vocalizations, such as laughter and screams. To explore acoustic distinctions across these vocal domains, we examined a diverse, cross-cultural dataset comprising over two hours of speech, singing, and nonverbal vocalizations. Our analysis reveals that speech tends to be lower in pitch and more tonal, characterized by regular phonation. In contrast, singing displays greater pitch variation, and nonverbal vocalizations exhibit extreme pitch ranges with frequent irregular phonation, often due to nonlinear phenomena, resulting in a harsh sound. While the evolution of complex supralaryngeal articulation has been essential for spectro-temporal modulation in speech, it has not significantly limited laryngeal source modulation. Nonverbal vocalizations, however, show minimal articulatory complexity, primarily featuring open vowels with rapid temporal modulation in the roughness range. We conclude that vocal source modulation is most effective for conveying affect, whereas vocal filter modulation primarily supports semantic communication.

OS2-6

Mice feel the beat of their songs

Aglaé Thieffry, Léo Perrier, Lorine Grangy, Floriane Fournier, Michael Greenfield, Nicolas Mathevon, Florence Levréro

Affiliation of presenting author:

ENES Bioacoustics Research Laboratory, CRNL, CNRS, Inserm, University of Saint- Etienne, Saint-Etienne, France.

The generation and perception of rhythmic sounds are fundamental features of human speech and music, yet their importance in animal communication remains largely unexplored. Although rhythmic patterns are present in various mammalian vocalizations, their functional significance for communication between individuals is poorly understood. Past research on rhythm perception has been limited to a few species and has primarily relied on conditioning tasks rather than observations of spontaneous behaviors. As a consequence, whether mammals use rhythmic patterns to convey information to conspecifics remains an open question. Mice, for instance, produce ultrasonic songs composed of alternating calls and silences, but the informational relevance of the rhythm of these songs for the receiving individuals is unclear. In this study, we show that African striped mice (Rhabdomys pumilio) perceive and respond to the temporal organisation of their ultrasonic songs. We conducted field playback experiments to examine their responses to natural and modified mouse songs with altered rhythm (modified note timing), syntax (rearranged note order), or both, using vocalizations from familiar and unfamiliar individuals. Independent alterations to rhythm or syntax triggered more pronounced behavioral responses than unmodified songs. However, simultaneous changes to both rhythm and syntax resulted in weaker reactions, suggesting that such extensively modified signals may lose their biological relevance. These results, which align with findings in pinnipeds, demonstrate that rhythm perception in mice is not merely a sensory trait but a critical component for decoding vocal signals.

OS2-7

Auditory dynamics during echolocation in freely-swimming trained and wild toothed whales.

Adam B. Smith, James Finneran, Mark Johnson, Jason Mulsow, Dorian Houser, Sara Ortiz, Danuta M. Wisniewska, Jonas Teilmann, Michael Ladegaard, Kristian Beedholm, Freja Jakobsen, Peter T. Madsen, Magnus Wahlberg

Affiliation of presenting author:

Sound Communication and Behavior Group, Department of Biology, University of Southern Denmark, DK

Echolocating animals rely on their sense of hearing to extract crucial information about their environment while actively moving through complex acoustic scenes. Empirical auditory measurements from echolocating toothed whales are limited to stationary experiments, which may not reflect natural biosonar behaviour. Thus, there is a significant knowledge gap in auditory reception and processing during mobile echolocation. We used a suction-cup-attached biologging tag with external surface electrodes to record auditory brainstem responses (ABRs) from four freely swimming, trained bottlenose dolphins and a wild harbor porpoise during echolocation-mediated prey capture. Tag-based ABRs were extracted by triggering on both the tagged individual's outgoing clicks and on its returning echoes during target approach. Clicking dynamics during target approach led to reduced peripheral auditory sensation of outgoing clicks while the peripheral auditory sensation of returning echoes was generally constant. Individuals often heard quiet echoes at comparable sensation levels to louder outgoing clicks and showed decreased sensation of their own clicks with increasing click rate as they approached a target. These results indicate that auditory sensory inputs dynamically change during target approach across multiple species. They also suggest the presence of a previously unspecified mechanism of gain control in these species for reducing the auditory sensation of self-produced clicks. Tag-based ABRs bridge the considerable knowledge gap stationary experimental arrangements and natural foraging behaviour. Moreover, the method is widely applicable to other toothed whale species, providing unprecedented insight into toothed whale echolocation.

OS2-8

Extending human vocal tract models to carnivores: Three models to validate ultrasound measurements of a harbor seal

Yannick Jadoul, Maria Goncharova; Colleen Reichmuth; W. Tecumseh Fitch; Andrea Ravignani; Bart de Boer

Affiliation of presenting author:

Department of Human Neurosciences, Sapienza University of Rome, Italy

The details of vocal production mechanisms in humans are well-understood, and vocal tract models closely match experimental observations. The extent to which such human-based models apply to non-human animal vocal tracts, which have evolved under different evolutionary pressures, remains an open question. We created three models of a carnivore's vocal tract inspired by the airborne vocalizations of a trained harbor seal and simultaneous ultrasound imaging of its tongue. A simple acoustic model considered the seal's vocal tract as a cone whose angle changes, and shows this shape's influence on formant frequencies. It provides the simplest possible baseline for the formant patterns of a typical carnivore vocal tract when only the jaw angle changes. We also applied a model (following Mermelstein, 1967) that maps the first two formants to an estimate of the vocal tract's cross-sectional area. The model-based reconstruction of our formant measurements shows how the estimates match the tongue shape we observed in the seal's ultrasound scans. Finally, a more complicated 3D finite element model simulated the acoustics of a vocal tract with the typical V-shaped carnivore mouth opening. We found the radial asymmetry of the V-shaped mouth opening did not significantly affect the first few formant frequencies. The model also confirmed that the effective vocal tract length is shorter than that of a cylindrical larynx-to-snout vocal tract model. Our findings highlight potentially important differences in vocal tract shape of seals relative to existing models, and provide a more appropriate foundation to evaluate the seal's articulatory capabilities.

OS2-9

Mating calls of prey interfere with the sensory systems of bats

Antoniya Hubancheva, Klaus Hochradel, Laura Stidahold, Katrin Dimitrova, Kaloyana Koseva, Holger R. Goerlitz

Affiliation of presenting author:

National Museum of Natural History, Bulgarian Academy of Sciences, 1000 Sofia, Bulgaria; Acoustic and Functional Ecology Group, Max Planck Institute for Biological Intelligence, 82319 Seewiesen, Germany

Mating signals attract the attention of potential mates but also serve as a guiding cue for predators. To avoid being eaten, courting animals employ diverse antipredator strategies that often compromise the time spent securing mating partners or the quality of the mating signal. Here, we test the hypothesis that conspicuous mating signals overwhelm the sensory systems of predators and therefore offer protection to their senders without compromising the attraction of mating partners. We explored the predator-prey interactions between the conehead katydid (Ruspolia nitidula) and the greater and the lesser mouse-eared bats (Myotis myotis and M. blythii) that eavesdrop on prey-generated signals. We (i) measured the soundscape of the foraging grounds of the bats; (ii) recorded the spatial distribution and 3D sound field generated by calling katydids; (iii) carried out a fine-scale diet analysis of wild bats; and (iv) performed controlled behavioural experiments where bats trained to catch tethered mealworms were exposed to recordings of R. nitidula. Our results show that despite the conehead katydids dominating the soundscape where the mouse-eared bats forage, they are only rarely consumed by M. blythii and never by M. myotis. Further, we demonstrate that R. nitidula choruses create an intense and broadband sound field that reduces the foraging success of both gleaning and aerial-hawking bats. This finding suggests that the mating signals of some katydid species have an antipredator function with the potential for acoustic dominance and sensory repercussions on entire ecosystems.

OS2-10

The social function of birdsong: new insights from zebra finches in the wild

Marc Naguib

Affiliation of presenting author:

Behavioural Ecology Group, Wageningen University, The Netherlands

Birdsong is an important model system for animal communication and typically functions in mate attraction and territory defense. The zebra finch *Taeniopygia guttata* is the most studied songbird under controlled laboratory conditions where they are commonly kept in pairs or same sex groups. Yet, in the wild zebra finches live in dynamic societies where males are not territorial, and pairs form long-term monogamous faithful bonds early in life, raising question on how they are socially and spatially organized and which role their song plays. To uncover their social organization and male singing we here integrated various bioacoustic measures with automated solar powered radio-tracking. Here I will present results on their singing behavior and on their movements and social organization. These findings on communication and social organization provide novel insights in the social life of birds and specifically the prime laboratory species in avian acoustics, the zebra finch.